

MAITHILI S

Generative AI Developer | LLM Systems Engineer | AI Research Engineer

Bangalore, India | maithilisaravanan05@gmail.com | +91-8431382181

LinkedIn: [Maithili Saravanan](#) | GitHub: [MaithyAnalyzent](#) | [Portfolio](#) | [HuggingFace](#)

PROFESSIONAL SUMMARY

Generative AI Developer and LLM Systems Engineer with hands-on experience building RAG pipelines, agentic AI architectures, and LLM evaluation frameworks. Proficient in Python, PyTorch, and transformer-based systems, with practical exposure to LLM inference, dataset curation for model finetuning, and evaluation-driven development. Experienced in deploying end-to-end AI applications that combine research-oriented experimentation with scalable product engineering.

Research interests: LLM Training & Inference Systems, Retrieval-Augmented Generation, AI Evaluation Frameworks, Autonomous AI Agents, Multi-Agent Reasoning, Trustworthy AI.

TECHNICAL SKILLS

AI / ML: RAG Pipelines, LLM Applications, Prompt Engineering, AI Agents, Multi-Agent Systems, LangChain, CrewAI, Langflow, Deep Learning, CNNs, UNet, AI Workflow Automation

Programming & Frameworks: Python, JavaScript, TypeScript, React.js, Next.js, FastAPI, HTML, CSS, Tailwind CSS

Tools & Platforms: PyTorch, TensorFlow, Scikit-learn, Gradio, Streamlit, ComfyUI, Git, GitHub, Cursor

Systems & Data: API Integrations, Experimentation Workflows, AI System Design, Data Visualization, Statistical Analysis

PROFESSIONAL EXPERIENCE

Software Engineer I (AI/ML) — Damco Solutions

Jun 2025 – Present

- Developing and optimizing LLM-powered pipelines using RAG architectures and LangChain workflows for enterprise-scale intelligent retrieval and reasoning systems.
- Designing agentic AI orchestration frameworks with focus on multi-step contextual inference, workflow automation, and retrieval quality improvement.
- Contributing to scalable LLM applications with emphasis on inference reliability, explainability, and evaluation-driven iteration.
- Collaborating across engineering teams on prompt optimization, AI system design, and performance benchmarking.

GCSE Educator (Mathematics & Science) — Self-Employed

Aug 2023 – Present

- Designed structured analytical learning systems for GCSE students specializing in Mathematics and Science.
- Developed customized problem-solving approaches and personalized study workflows.
- Conducted remote mentoring sessions focused on conceptual understanding and critical thinking.

Data Annotator (Freelance) — RWS Group

Feb 2024 – Oct 2024

- Curated and annotated high-quality training datasets for LLM and ML model development, pretraining, and finetuning workflows.
- Established annotation consistency standards to improve training data quality and reliability for generative model pipelines.

Web Developer — TechVolt Software Pvt. Ltd.

Jul 2021 – Sep 2021

- Built responsive web applications and optimized frontend user experiences.

- Contributed to deployment workflows and application functionality improvements.

EDUCATION

MBA in Data Analytics — Manipal University Jaipur (Online) | 2023 – 2025

BCA in Data Science — Amrita Vishwa Vidyapeetham | 2020 – 2023

100xEngineers GenAI Cohort | Jun 2024 – Dec 2024

ACHIEVEMENTS

- Finalist — 100xEngineers Generative AI Buildathon
- Poster Presentation — "Retrained UNet Neural Network for Glaucoma Detection" at ICCHA 2023
- 2nd Prize — Hackathon, Hindustan College, Mysore
- Participant — BuildSpace Nights & Weekends Season 5
- Event Coordinator — BrainZee Coding Event, Sathgamaya 2022

RESEARCH INTERESTS

LLM Training & Inference Systems, AI Search Systems, Trustworthy AI, Autonomous AI Agents, Retrieval-Augmented Generation, AI Evaluation Frameworks, Multi-Agent Reasoning Systems

LANGUAGES

English, Tamil, Kannada, Hindi, Telugu, Malayalam